WTW 158 - **WORKSHEET ON DIFFERENTIATION -2016**

Find the derivatives of the following functions. Don't simplify your answers.

1. $f(x) = x^2 e^{4x}$

2. $f(\theta) = \dfrac{\tan 2\theta}{1 + \sec(\theta^2)}$ (Use the quotient rule)

3. $f(t) = t^3 + e^t + 3^t$

4. $f(t) = (1 - 4t)^3 + e^{1-5t} + 3^{1-6t}$

5. $f(x) = 5\sin(x^4)$

6. $f(x) = \sin^4 5x$

7. $f(t) = e^{\cos 5t}$

8. $f(t) = \cos(e^{5t})$

9. $f(u) = \ln(1 - \sqrt[3]{u})$

10. $f(u) = \sqrt[3]{1 - \ln 2u}$

11. $f(x) = \sec(\ln(2x + 6))$

12. $f(t) = \ln(\sec(3x + 7))$

13. $y = \operatorname{cosec}(3x^2)$

14. $y = 3^{\operatorname{cosec} x^2}$

15. $f(x) = \tan^{-1}(e^{8x})$ (Notation: $\tan^{-1} = \arctan = \text{bgtan}$)

16. $f(t) = \sin^{-1}(1 - 9t^4)$ (Notation: $\sin^{-1} = \arcsin = \text{bgsin}$)

17. $x^3 \sin y + xy^4 = y + e$

18. $f(t) = t^{\cos 2t}$

19. Find the equation of the tangent line to the graph of $f(x) = 2e^{1-x}$ where $x = 1$.

20. Find all x values in the domain of $f(x) = \dfrac{\ln 2x}{x^3}$ where you can draw a tangent line parallel to the x –axis